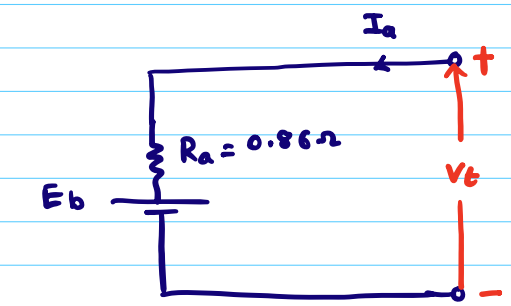
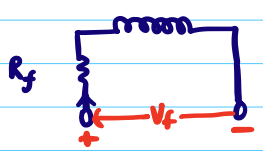
EE2005 : TUTORIAL - 4 on DC Machines-I

1. A separately excited DC motor has back-emf constant, $k\phi = 1.44\,\text{V/rad/s}$, $R_a = 0.86\,\Omega$. When it is operating at $150\,\text{rad/s}$, the armature current is $I_a = 40\,\text{A}$. The terminal voltage V_t is held constant under all conditions
   (a) Compute V_t
   (b) Compute the no-load speed in rad./s (Neglect core and mechanical losses)



$\Rightarrow V_t = E_b + I_a R_a$



$\omega = 150$ rad/sec

$I_a = 40$ A

$K\phi = 1.44$ Volt/rad/sec

We know $E_b = K\phi\,\omega = \dfrac{\text{Volt}}{\text{rad/sec}} \times \text{rad/sec}$

a)

After applying KVL

$V_t = E_b + I_a R_a$

$= K\phi\,\omega + I_a R_a$

$$V_t = 1.44 \times 150 + 40 \times 0.86$$

$$V_t = 250.4\,V$$

b) @ No-load $I_a = 0\ A$

$$\Rightarrow \quad V_t = E_b + I_a R_a$$

$$V_t = K\phi\,\omega + 0$$

$$\omega = \frac{V_t}{K\phi}$$

$\therefore$ speed at no load $\omega_{nl} = \dfrac{V_t}{K\phi}$

$$= \frac{250.4}{1.44}$$

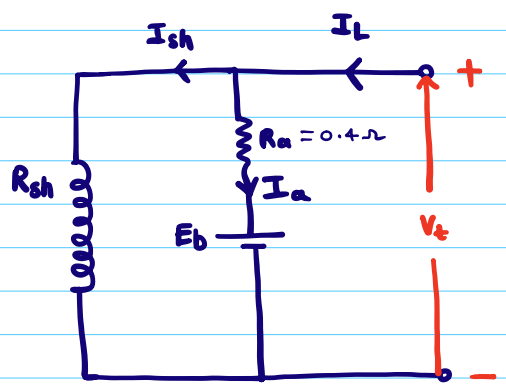
$$\Rightarrow \quad \omega_{nl} = 173.889\ rad/sec$$

or

$$N_{nl} = \omega_{nl} \times \frac{60}{2\pi}$$

$$N_{nl} = 1660.5166\ RPM$$

2. A 200 V DC shunt motor has an armature resistance of 0.4 Ω. When operated from the rated supply voltage at full load, the motor runs at 500 rpm and draws an armature current of 25 A. If a resistance of 1.6 Ω is placed in series with the armature, find the speed at (i) full load torque (ii) three times full load torque. Neglect core and mechanical losses.



@ Full load ,

$$V_t = 200\,V$$

Armature draws 25 A

i.e., $I_a^f = 25\,A = I_{a1}$

$N_{full} = 500\,RPM = N_1$

$$\omega_{full} = \frac{2\pi N_{full}}{60}$$

$$= \frac{2\pi * 500}{60}$$

$$= 52.3599 \text{ rad/sec}$$

ω_{m1} : let speed of motor at full load (rad/sec)

$\omega_{m1} = \omega_{full} = 52.3599$ rad/sec

Now KVL

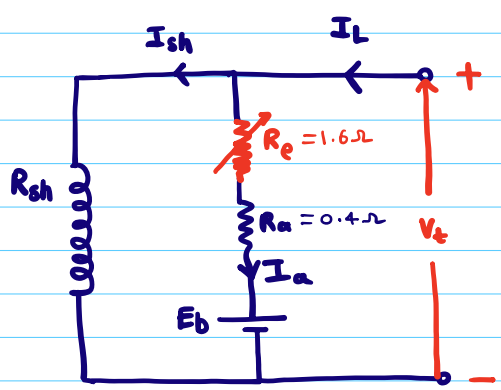
$$V_t = E_b + I_a R_a$$

$$K_e \omega = V_t - I_a R_a$$

$$\boxed{K_e = \frac{V_t - I_a R_a}{\omega}}$$

$$K_e = \frac{V_t - I_a^{f} R_a}{\omega_{m1}} = \frac{200 - 25 \times 0.4}{52.3599}$$

$$K_e = 3.6287 \text{ V/rad/sec}$$



Re : Extra Resistance in series with the armature Resistance.

* Simple Implementation
* less efficient

$$\omega_m = \frac{V_t}{K_c \phi} - \frac{R_a + R_c}{(K_c \phi)^2} T$$

(i) when R_c is added

T_2 = Full load Torque (say)

$$E_{b_2} = V_t - I_{a_2}(R_a + R_c)$$

$$E_{b_1} = V_t - I_{a_1} R_a$$

$E_b \propto \omega_m$

$$\omega_{m_2} = \frac{E_{b_2}}{E_{b_1}} \omega_{m_1}$$

$I_{a_2} = I_{a_1}$ = Rated Armature current

$$\omega_{m_2} = \frac{V_t - I_{a_2}(R_a + R_c)}{V_t - I_{a_1} R_a} \times \omega_{m_1}$$

or

$$N_2 = \frac{V_t - I_{a_2}(R_a + R_c)}{V_t - I_{a_1} R_a} \times N_1$$

$$= \frac{200 - 25(0.4 + 1.6)}{200 - 25 \times 0.4} \times 500$$

$$N_2 = 394.7368 \text{ RPM}$$

in rad/sec

$$\omega_{m2} = \frac{2\pi N_2}{60}$$

$$\omega_{m2} = 41.3367 \text{ rad/sec}$$

(ii) $T_3 = 3 * T_2$

we know

$$T \propto I$$

$$\frac{T_3}{T_2} = \frac{I_{a3}}{I_{a2}}$$

$$I_{a3} = \frac{T_3}{T_2} I_{a2}$$

$$= 3 \times 25$$

$$I_{a3} = 75 \text{ Amps}$$

$$N_3 = \frac{V_t - I_{a3}(R_a + R_c)}{V_t - I_{a1} R_a} \times N_1$$

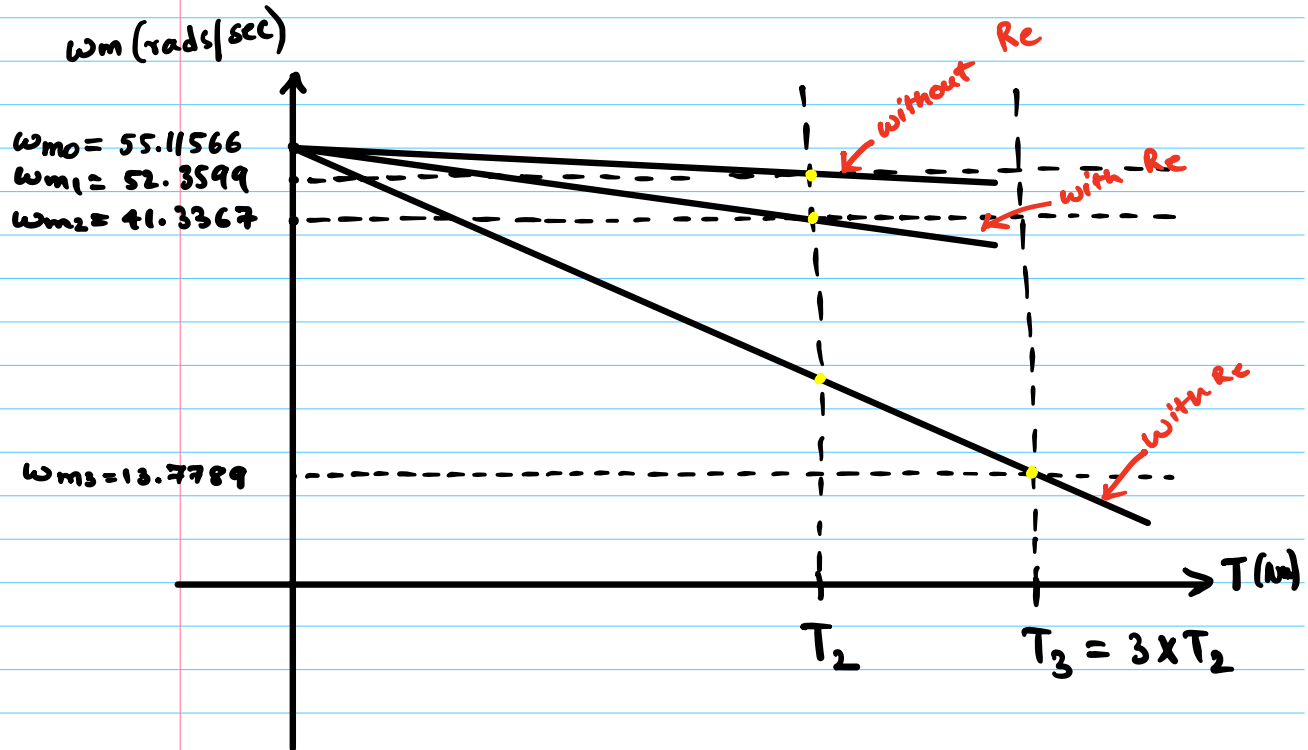
$$= \frac{200 - 75\,(0.4 + 1.6)}{200 - 25 \times 0.4} \times 500$$

$$N_3 = 131.5789 \text{ RPM}$$

or

$$\omega_{m3} = \frac{2\pi N_3}{60}$$

$$\omega_{m3} = 13.7789 \text{ rad/sec}$$



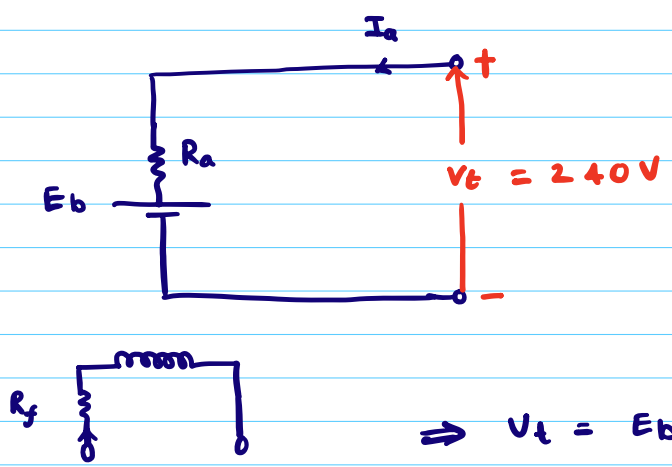
@ No load:

$$\omega_{mo} = \frac{V_t}{k_e} = 55.11566 \text{ rad/sec}$$

3. A separately excited DC motor is rated for a full load torque of 100 Nm. For an applied voltage of 240 V, the no-load and full load speeds of the motor are 1595 rpm and 1195 rpm respectively, find

   (a) (i) The back-emf constant k_e of the machine
   (ii) The armature resistance R_a

   (b) If the applied voltage and the field of the machine are reduced to 83.33% and 66.66% of the previous values respectively, find
   (i) The corresponding no load speed
   (ii) The corresponding full load speed



$\Rightarrow V_t = E_b + I_a R_a$

$T_f = 100$ Nm

$N_0 = 1595$ RPM $\Rightarrow \omega_{m0} = 167.028$ rad/sec

$N_1 = 1195$ RPM $\Rightarrow \omega_{m1} = 125.140$ rad/sec

$\phi \neq$ constant

$$V_t = E_b + I_a R_a$$

$$R_a = ?$$

a. (i) $V_t = E_b$ at No-load

$$= K_e \omega_{m0}$$

$$K_e = \frac{V_t}{\omega_{m0}}$$

$$= \frac{240}{\frac{2\pi 1595}{60}}$$

$$K_e = 1.43688 \ V/rad/sec$$

(ii) $V_t = E_b + I_a R_a$

At full load condition

$$V_{t_1} = E_{b1} + I_{a1} R_a$$

$$V_{t_1} = K_e \times \omega_{m1} + I_{a1} R_a$$

$$I_{a1} R_a = V_{t_1} - K_e \times \omega_{m1}$$

$= 240 - 1.43688 \times 125.14$

$= 60.188\ V$

$I_{a1} R_a = 60.188\ V$

To find I_{a1} (full load current)

we know $T = K_t I_a = K_e I_a$

$T_{full} = K_e I_{a1}$

Given, $T_{full} = 100\ N\,m$

$$I_{a1} = \frac{T_{full}}{K_e}$$

$$= \frac{100}{1.43688}$$

$I_{a1} = 69.595$ amps

$$R_a = \frac{60.188}{I_{a1}}$$

$$= \frac{60.188}{69.595}$$

$R_a = 0.8648\ \Omega$

b.(i) $V_{t2} = V_{t1} \times 83.33\%$

$= 240 \times 83.33\%$

$= 199.992\ V$

But $\phi_2 = \phi_1 \times 66.66\%$

$E_b \propto K_c \phi$

$$\frac{K_{c2}}{K_{c1}} = \frac{66.66\%\ \phi_1}{\phi_1} = 0.6666$$

$K_{c2} = 0.9578\ V/rad/sec$

$$\omega_{m0} = \frac{V_{t2}}{K_{c2}} = \frac{199.992}{0.9598}$$

$\omega_{m20} = 208.7975$ rads/sec

or $N_{20} = \omega_{m20} \frac{60}{2\pi}$

$N_{20} = 1993.8696$ RPM

(ii) $T \propto K_t \phi I_a$

$$I_{a2} = \frac{K_{c1}}{K_{c2}} I_{a1}$$

$$= \frac{1.4368}{0.9578} \times 69.59$$

$$I_{a2} = 104.4029 \text{ Amps}$$

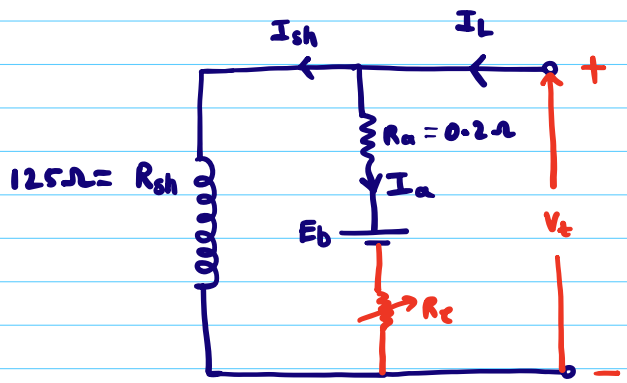
$$\omega_{m2} = \frac{V_{t2} - I_{a2} R_a}{K_{c2}}$$

$$= \frac{199.992 - 104.4029 \times 0.8648}{0.9578}$$

$$\omega_{m2} = 114.53 \text{ rad/sec}$$

$$N_2 = 1093.69 \text{ RPM}$$

4. A 10 kW, 250 V, 1200 rpm DC shunt motor has a full load efficiency of 80%. The field and armature resistances are 125 Ω and 0.2 Ω respectively. The speed of the motor is to be reduced to 75% with load torque remaining constant. What resistance should be inserted in the armature circuit. Neglect stray losses.



Given $P_o = 10\,kW$ (Shaft Output Power)

$$P_{mech.dev} = E_b \times I_a \neq P_o$$

$$P_o = P_{mech.dev} - \text{losses}$$

$$\%\,\eta = 80\,\%$$

$$\frac{P_o}{P_i} = 0.8$$

$$P_i = V_t \times I_L = \frac{P_o}{0.8}$$

$$I_L = \frac{1}{V_t} \times \frac{P_o}{0.8}$$

$$I_L = \frac{1}{250} \times \frac{10000}{0.8}$$

$$I_L = 50 \text{ Amps}$$

$$I_{sh} = \frac{V_t}{R_{sh}}$$

$$= \frac{250}{125}$$

$$I_{sh} = 2 \text{ Amps}$$

$$I_a = I_L - I_{sh}$$

$$= 50 - 2 = 48 \text{ Amps}$$

$$I_a = 48 \text{ Amps}$$

$$K_c = \frac{V_t - I_a R_a}{\omega_m}$$

$$\text{given } \omega_m = 1200 \times \frac{2\pi}{60}$$

$$\Rightarrow K_c = \frac{250 - 48 \times 0.2}{1200 \times \frac{2\pi}{60}}$$

$$K_c = 1.913 \text{ V/rad/sec}$$

with T constant

$$E_b = k_c \omega_m$$

$$E_b \propto \omega_m$$

$$\frac{E_{b2}}{E_{b1}} = \frac{\omega_{m2}}{\omega_{m1}}$$

Given $\omega_{m2} = 75\% \, \omega_{m1}$

$$\Rightarrow \quad \frac{E_{b2}}{E_{b1}} = \frac{0.75 \, \omega_{m1}}{\omega_{m1}}$$

$$\text{or} \quad \frac{V_t - I_a (R_a + R_c)}{V_t - I_a R_a} = 0.75$$

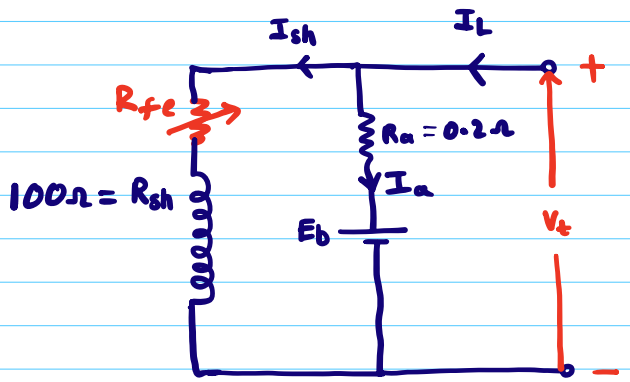
$$V_t - I_a (R_a + R_c) = 0.75 (V_t - I_a R_a)$$

$$R_c = \frac{V_t - 0.75 (V_t - I_a R_a) - I_a R_a}{I_a}$$

$$R_c = \frac{250 - 0.95(250 - 48 \times 0.2) - 48 \times 0.2}{48}$$

$$R_c = 1.25208 \ \Omega$$

5. A 250 V DC shunt motor has an armature resistance of 0.2 Ω and shunt field resistance of 100 Ω . It takes 60 A when running at a speed of 600 rpm. If the speed is to be increased to 800 rpm, Find the resistance of the field regulator to be inserted. Assume the magnetisation characteristic to be a straight line and the load torque remains constant.



Given

$$I_L = 60 \text{ Amps}$$

$$N = 600 \text{ RPM}$$

$$I_{sh} = \frac{V_t}{R_{sh}}$$

$$= \frac{250}{100}$$

$$I_{sh} = 2.5 \text{ A}$$

By KCL, $I_a = I_L - I_{sh}$

$$I_a = 60 - 2.5$$

$$I_a = 57.5 \text{ Amps}$$

$$E_b = V_t - I_a R_a$$

$$= 250 - 57.5 \times 0.2$$

$$E_b = 238.5\ V$$

$$K_c\, \omega_m = 238.5\ V$$

$$K_c = \frac{238.5}{\omega_m}$$

given $N = 600$ rpm

$$\omega_m = \frac{2\pi 600}{60}$$

$$\omega_{m1} = 62.8318 \text{ rad/sec}$$

$$K_{c1} = \frac{238.5}{62.8318}$$

$$K_{c1} = 3.79584 \text{ V/rad/sec}$$

$K_t I_a \propto$ constant

$$\omega_{m2} = 800 \times \frac{2\pi}{60}$$

$$\omega_{m2} = 83.7758 \text{ rad/sec}$$

$$I_{a2} = \frac{V_t - K_{e2}\omega_{m2}}{R_a}$$

$$I_{a2} = \frac{V_t}{R_a} - \frac{K_{e2}\omega_{m2}}{R_a} \quad — (a)$$

$T = K_e I_a$

$I_{a1} = 57.5A$

$K_{e1} = 3.79584$ V/rad/sec

$$\frac{T_1}{T_2} = \frac{k_{e1} I_{a1}}{k_{e2} I_{a2}}$$

$$K_{e2} = \frac{I_{a1}}{I_{a2}} k_{e1} \quad — (b)$$

put (b) in (a)

$$I_{a2} = \frac{V_t}{R_a} - k_{e1} \frac{I_{a1}}{I_{a2}} \frac{\omega_{m2}}{R_a}$$

$$I_{a2}^2 - \frac{V_t}{R_a} I_{a2} + k_{c1} I_{a1} \frac{\omega_{m2}}{R_a} = 0$$

$I_{a2} = 1171.991\ A$ — NOT Feasible value

or $\boxed{I_{a2} = 78.008\ A}$ → Feasible value

With $I_{a2} = 78.008\ A$,

the back emf constant is

$$K_{c2} = \frac{V_t - I_{a2} R_a}{\omega_{m2}}$$

$$= \frac{250 - 78.008 \times 0.2}{83.775}$$

$$K_{c2} = 2.7979\ V/rad/sec$$

Another way to calculate Back EMF

Constant K_{e_2} :

Re-writting (b) :

$$I_{a_2} = I_{a_1} \times \frac{K_{e_1}}{K_{e_2}} \rightarrow (c)$$

put (c) in (a) :

$$\frac{K_{e_1}}{K_{e_2}} I_{a_1} = \frac{V_t}{R_a} - K_{e_2} \frac{\omega_{m_2}}{R_a}$$

$$K_{e_2} \frac{\omega_{m_2}}{R_a} - K_{e_2} \frac{V_t}{R_a} + K_{e_1} I_{a_1} = 0$$

with $K_{e_1} = 3.79584$ V/rad/sec being known

solve above equ for K_{e_2}

$$K_{e_2} = 2.797 \text{ V/rad/sec}$$
$$= 0.1862 \text{ V/rad/sec}$$

put K_{e_2} in $I_{a_2} = \frac{V_t - K_{e_2} \omega_{m_2}}{R_a}$

	$K_{e1} = 2.7979$	$K_{e2} = 0.1862$
I_{a2}	78.01838 A	1172.0047 A
No-load speed (RPM) $\left(\frac{V_t}{K_e}\right)$	853.2557 RPM	12821.2897 RPM

* So with $K_{e2} = 0.1862$ V/rad/sec

No-load speed is 12821.2897 RPM

and armature current is 1172.0047 A

which are not feasible.

To find R_{fc}, extra resistance inserted in field Circuit.

R_{sh1} : Field resistance initially

R_{sh2} : New field resistance after adding/inserted extra to R_{sh1} $= R_{sh1} + R_{fc}$

$$K\phi = K_c \propto I_f \propto I_{sh} = \frac{V}{R_{sh}}$$

$$\Rightarrow K_c \propto \frac{1}{R_{sh}}$$

$$\frac{K_{c1}}{K_{c2}} = \frac{R_{sh2}}{R_{sh1}}$$

$$R_{sh2} = R_{sh1} \times \frac{K_{c1}}{K_{c2}}$$

$R_{sh1} = 100\,\Omega$

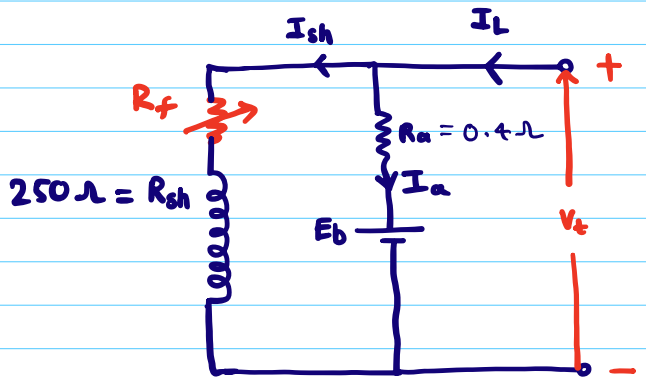
$$= 100 \times \frac{3.7958}{2.7979}$$

$$= 135.666\,\Omega$$

$$R_{sh2} = R_{fc} + R_{sh1}$$

$$R_{fc} = 35.6665\,\Omega$$

6. A 250 V shunt motor develops a total torque of 105 Nm and takes 36 A at 750 rpm. The armature and shunt field resistances are 0.4 Ω and 250 Ω respectively. If the speed is to be increased to 1000 rpm, determine the percentage reduction of the field and additional resistance to be inserted in the field circuit. Total toque developed at 1000 rpm is 75 Nm. Assume the magnetisation characteristic to be a straight line.



when motor runs @ 750 RPM :

$$I_L = 36\ A$$

$$I_{sh} = \frac{250}{250}$$

$$= 1\ A$$

$$I_{a1} = I_L - I_{sh} \quad \text{(by KCL)}$$

$$= 36 - 1$$

$$I_{a1} = 35\ \text{Amps}$$

$$\omega_{m1} = \frac{2\pi\ 750}{60}$$

$$\omega_{m1} = 78.5398\ \text{rad/sec}$$

$$V_t = E_{b1} + I_{a1} R_a$$

$$= K_{c1} \omega_{m1} + I_{a1} R_a$$

$$K_{c1} = \frac{V_t - I_a R_a}{\omega_{m1}}$$

$$K_c = 3.0048$$

$$T_{e1} = K_{t1} I_{a1}$$

$$K_{t1} = \frac{T_{e1}}{I_{a1}}$$

$$K_{t1} = \frac{105}{35} = 3$$

$\therefore K_{c1} \approx K_{t1}$

$K_{c1} = 3.0048$ V/rad/sec

To Calculate K_{c2} for increasing speed to 1000 RPM from 750 RPM

$$\omega_{m2} = \frac{2\pi \, 1000}{60}$$

$$\omega_{m2} = 104.7197 \text{ rad/sec}$$

$$I_{a2} = \frac{V_t}{R_a} - \frac{K_{c2} \omega_{m2}}{R_a} \rightarrow (a)$$

Electromagnetic Torque developed (T_e)

$$T_e = K_t I_a = K_c I_a$$

$$\frac{T_{e2}}{T_{e1}} = \frac{K_{c2}}{K_{c1}} \frac{I_{a2}}{I_{a1}}$$

$$\frac{75}{100} = \frac{K_{c2}}{K_{c1}} \times \frac{I_{a2}}{I_{a1}}$$

$$I_{a2} = I_{a1} \frac{k_{e1}}{k_{e2}} \times \frac{75}{100}$$

$$I_{a1} \frac{k_{e1}}{k_{e2}} \times \frac{75}{100} = \frac{V_t}{R_a} - \frac{k_{e2}\, \omega_{m2}}{R_a}$$

$$\frac{\omega_{m2}}{R_a} k_{e2}^2 - \frac{V_t}{R_a} k_{e2} + \frac{75}{105} k_{e1} I_{a1} = 0$$

Solve k_{e2}

$k_e = 2.260382$ V/rad/sec

$= 0.12694$ V/rad/sec

	k_{e2}	
	2.260382	0.12694
I_{a2}	33.233 A	591.767 A
No load speed $\left(\frac{V_t}{k_e}\right)$	1056.1595 RPM	18806.713 RPM

Not feasible

So,

$K_{c2} = 2.260362 \text{ V/rad/sec}$

New field resistance after adding extra resistance in field circuit.

$$R_{sh2} = R_{sh1} \times \frac{K_{c1}}{K_{c2}}$$

$$= 250 \times \frac{3.0048}{2.260362}$$

$$R_{sh2} = 332.333$$

$$R_{fe} = R_{sh2} - R_{sh1}$$

$$= 332.33 - 250$$

$$R_{fe} = 82.3333 \ \Omega$$

% Field Reduction Calculation :

Approach 1 :

$$\% \text{ Field reduction} = \left| 1 - \frac{k_{e2}}{k_{e1}} \right| \times 100$$

$$= 0.2477 \times 100$$

$$\% \text{ Field reduction} = 24.77\% \text{ reduction}$$

Approach 2 :

$$I_{sh2} = \frac{R_{sh1}}{R_{sh2}} I_{sh1} \qquad I_{sh1} = 1\,A$$

$$= \frac{250}{331.33} \times 1$$

$$I_{sh2} = 0.75226\,A$$

$$\% \text{ Field red} = \left| 1 - \frac{I_{sh2}}{I_{sh1}} \right| \times 100$$

$$= 24.77\% \text{ Reduction}$$

Approach 3 :

$$\% \text{ Field Red} = \left| 1 - \frac{R_{sh1}}{R_{sh2}} \right| \times 100$$

$$= \left| 1 - \frac{250}{332.33} \right| \times 100$$

$$= 0.2477 \times 100$$

$$= 24.77\% \text{ Reduction}$$